

Associated content

Laser fragmentation of colloidal gold nanoparticles with high-intensity nanosecond pulses driven by a single-step fragmentation mechanism with a defined educt particle-size threshold

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Size distribution of the educt particles after PLAL

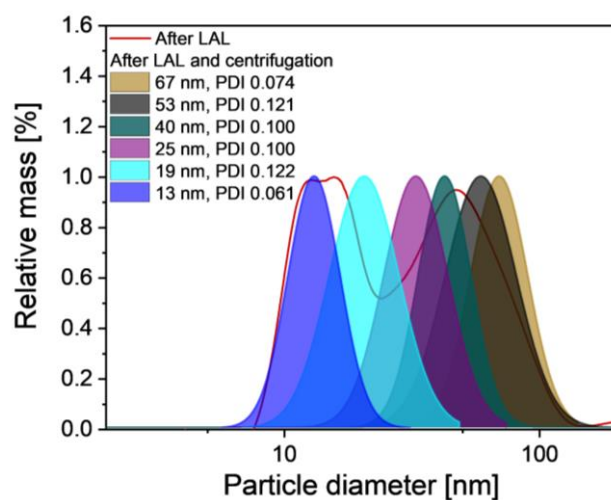


Figure S 1 Relative mass distribution of gold nanoparticles after LAL and fractionated centrifugation. Measured using ADC.

Calculations of the flow behavior of the free liquid jet

The use of a free liquid jet during LFL is an unconventional method that is rarely reported in the literature. Fragmentation typically takes place in a cuvette in which the liquid is stirred, which results in turbulent flow conditions.

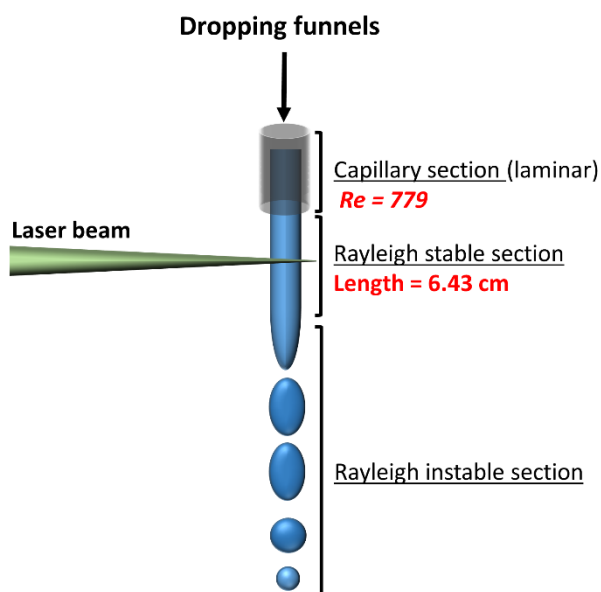


Figure S 2: Schematic diagram of fluid dynamic observation of the free liquid jet

The free liquid jet is characterized in particular by its laminar flow behavior. The Reynolds number inside the nozzle was calculated to be $Re = 779$. Once the liquid exits the capillary, the gravity, flow velocity v and the surface tension σ of the medium determine its subsequent fate.

$$v > v_{Min} = 2 \cdot \sqrt{\frac{\sigma}{r_{Jet} \cdot \rho}} \quad \text{equation 1}$$

Starting from a surface tension of 0.72 N/m and a density ρ of water of 997.04 kg/m³, the minimum discharge velocity is $v_{Min} = 0.73$ m/s. Since the present flow velocity of 0.63 m/s is lower, decomposition of the liquid jet into droplets is to be expected. In our experiments, the laser pulse irradiates the continuous area before dripping begins (Figure S 2). The drip-off length L of this range can be calculated by multiplying the decay time t by the flow velocity v_{Jet} :

$$L = t \cdot v_{Jet} = \frac{v_{Jet}}{\mu_{opt}} \cdot \ln\left(\frac{r_{Jet}}{\delta_0}\right). \quad \text{equation 2}$$

Therein, the ratio $\ln\left(\frac{r_{Jet}}{\delta_0}\right)$ depends on the Ohnesorge number Oh , and can be calculated according to Grant and Middleman as follows [1]:

$$\left(\frac{r_{Jet}}{\delta_0}\right) = -2.66 \cdot \ln(Oh) + 7.68 \quad \text{equation 3}$$

The Ohnesorge number can be calculated $\left[Oh = \frac{\sqrt{We}}{Re} = \frac{\eta}{\sqrt{L \cdot \rho \cdot \sigma}}\right]$. The optimal Rayleigh growth rate for non-viscous jets can be calculated as follows [1]:

$$\mu_{opt} = 0.97 \cdot \sqrt{\frac{\sigma}{8 \cdot \rho \cdot r_{Jet}^3}} \quad \text{equation 4}$$

Finally, length of the continuous section, determining the working range is approximated by equation 1 and results in $L = 6.4$ cm. This was in good agreement with experimental observation.

In our experimental setup the working distance was set at $L = 1$ cm. To estimate whether the liquid jet is dominated by inertial forces (flow rate) or gravitational forces we calculate the Froude number according to:

$$Fr^2 = \frac{v_{Jet}}{\sqrt{g \cdot L}} \quad \text{equation 5}$$

Under these conditions we obtain a Froude number of $Fr^2 = 4$. Hence the liquid jet is already strongly dominated by gravitational forces such that we assume plug flow-like conditions and similar residence time of nanoparticles within the irradiated area of the liquid jet.

Consideration of the absorption cross section of gold nanoparticles as a function of the laser wavelength

To decide which wavelength to use for fragmentation, we calculated the absorption cross sections of gold nanoparticles of different sizes using methods adapted from the literature. For the calculation, we used the freely available software MiePlot [2]. The absorption cross section is clearly dependent on the particle size and the wavelength used.

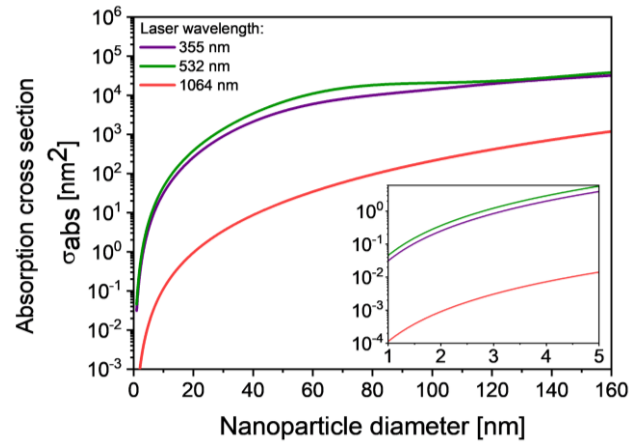


Figure S 3: Absorption cross section of gold nanoparticles of different sizes, calculated using MiePlot [1]. The absorption cross section depends on the excitation wavelength. In the calculations, we used optical data according to Johnson and Christy [2].

The absorption cross section increases with increasing particle size. In addition, it is highest when illuminated with light of 532-nm wavelength. Measured by these values, the laser fragmentation carried out using an excitation wavelength of 532 nm.

Comparison of the quality of three different methods for obtaining particle size distributions of particles after fragmentation

To verify the quality of the ADC in the measurement of particles down to 2 nm diameter, we compared the size distributions with those determined using AUC and TEM (Figure S4). The TEM and ADC distributions show very high comparability, both in the position of the peak maximum ($\pm 0.75\%$) and the width ($\pm 10.6\%$) of the distribution. The AUC data show a considerably higher resolution that enables the identification of several peaks. However, such a high resolution is not necessary for this study, as the primary comparison is between different distributions. In Figure 4 the data of the AUC are fitted without consideration of the high resolution to show a fair comparison to the ADC data. Based on this comparison, we assess the ADC as suitable measurement method for the measurement of particles in the range between 2-5 nm.

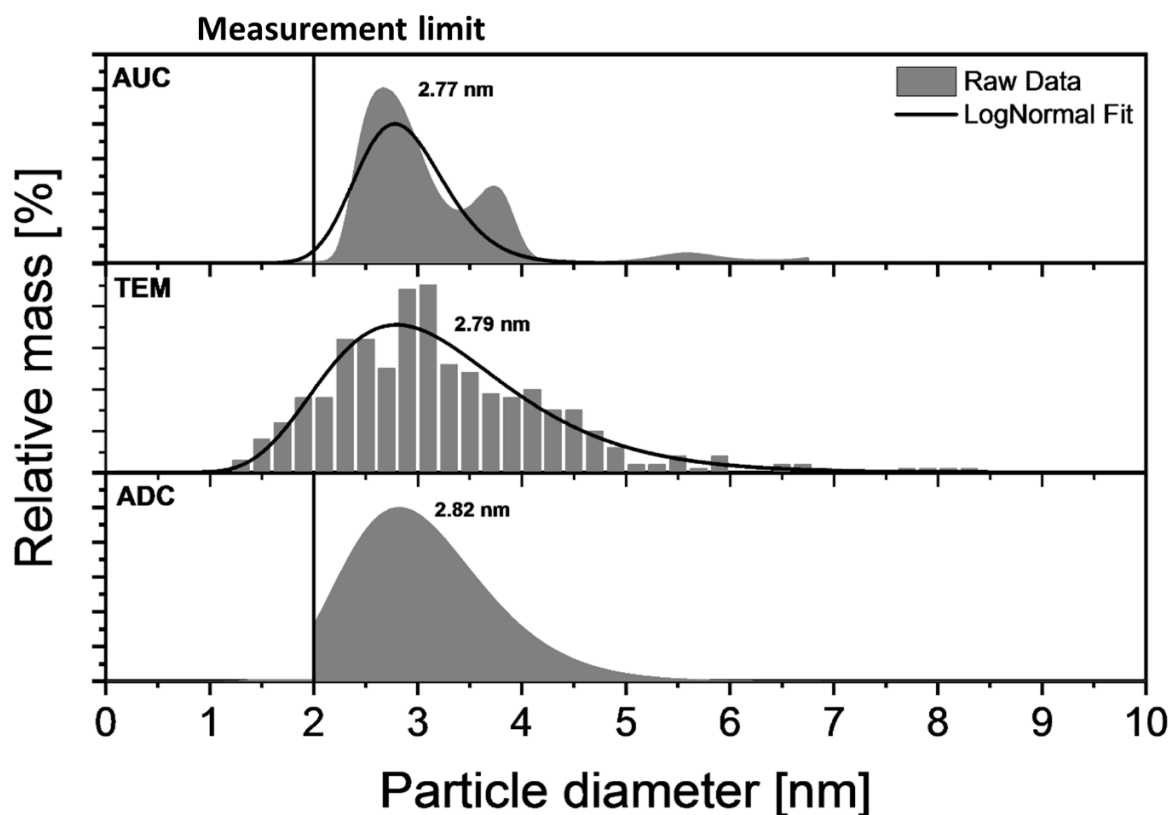


Figure S 4: Comparison of size distributions obtained from TEM, AUC, and ADC

Theoretical consideration of the temperatures in a nanoparticle by LFL

The amount of heat absorbed by a single nanoparticle and its respective temperature increase induced by a single laser pulse depends on the applied laser fluence, absorption efficiency at a given laser wavelength, and the heat-loss kinetics of the liquid. Baffou and co-workers provided a MatLab script that enabled us to estimate the deposited heat and resulting temperature increase of gold nanoparticles based on the Mie theory and heat transfer equations [3]. We performed these calculations based on the laser fluences and pulse durations applied in our experiments, the results of which are shown in Figure S 5. The dotted line in Figure S 5 indicates the energy required to heat the particle to boiling temperature ($T_b = 3129\text{ K}$), considering the heat capacity ($129\text{ J kg}^{-1}\cdot\text{K}^{-1}$) and the melting enthalpy (12.7 kJ mol^{-1}) [4]. Above this line, the AuNPs absorb enough energy to be partially evaporated. Once the mass-specific thermal energy of the AuNP also exceeds the evaporation enthalpy of gold (324 kJ mol^{-1}), complete evaporation of the AuNPs can be assumed (indicated by the dashed line in Figure S 5). Accordingly, the onset of fragmentation can be considered to occur at this point for the ‘ns-low-RR’ and ‘ns-high-RR’ lasers.

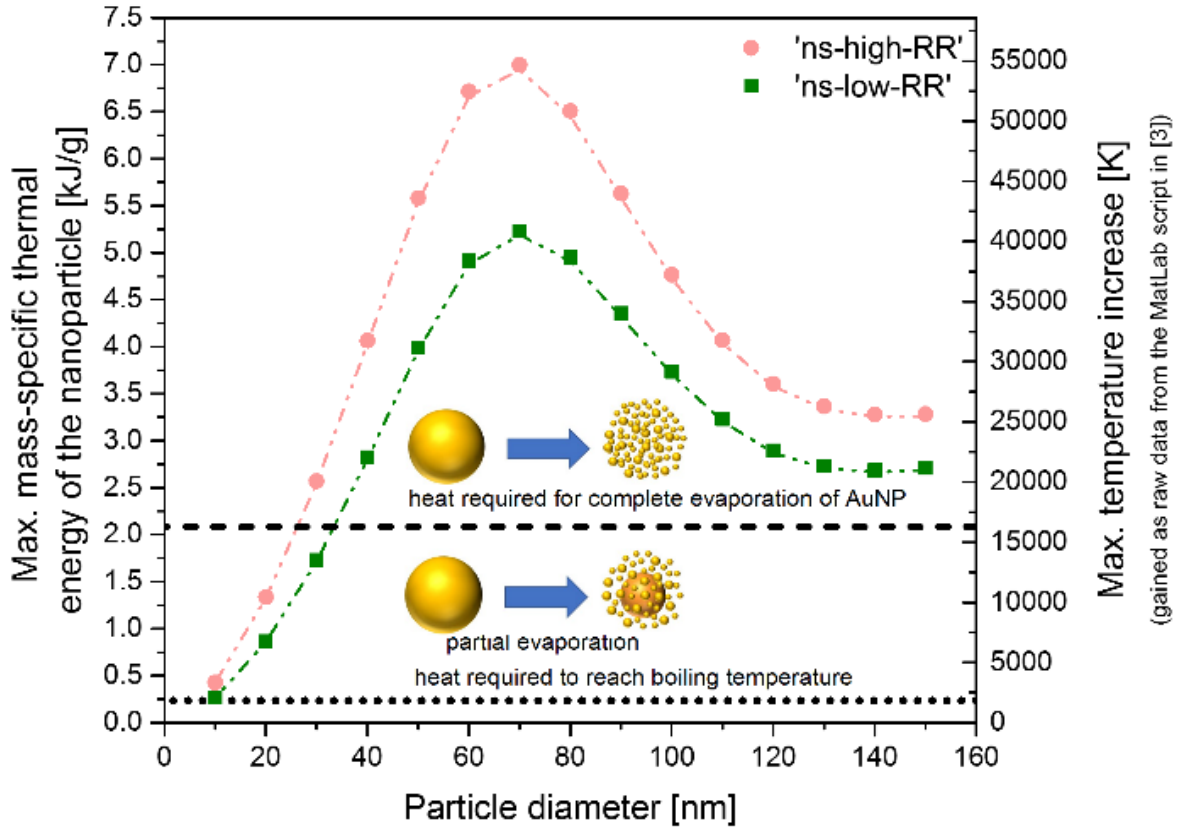


Figure S 5: Mass-specific thermal energy (left axis) absorbed by the AuNP during laser irradiation, as calculated from the virtual temperature increase (right axis) determined by the MATLAB script from ref. [5]. Data refers to the center of a gold nanoparticle, calculated for single-pulse experiments with the lasers used in this study.

For the sake of transparency, on the right hand side of Figure S 5, we plotted the maximal temperature increase obtained as raw data from the MatLab script (from ref. [5]). Note that within the model developed by Baffou and co-workers, the temperature increase is calculated solely from the absorbed energy and heat capacity, while neglecting melting and evaporation events [5]. Hence, once the melting temperature is exceeded, the temperature increase from the MatLab script no longer depicts the actual nanoparticle temperature (as melting and later evaporation enthalpy are not considered), but is rather a direct measure of the thermal energy stored in the AuNP. Therefore, we used the thermal energy throughout in our consideration of

the onset and completeness of particle evaporation, rather than the temperature obtained from the MatLab script.

Note that the calculation used by Baffou and co-workers (from ref. [5]) provides a ‘best-case-scenario’ prediction of the thermal energy absorbed by the AuNP under laser pulse irradiation. That is, as the refractive index of Au was kept constant throughout the whole calculation, no bleaching of surface plasmon resonance as reported by Werner et al. or subsequent drop in absorption efficiency with increasing temperature was neglected [6].

Absorption spectra of nanoparticles before and after LFL

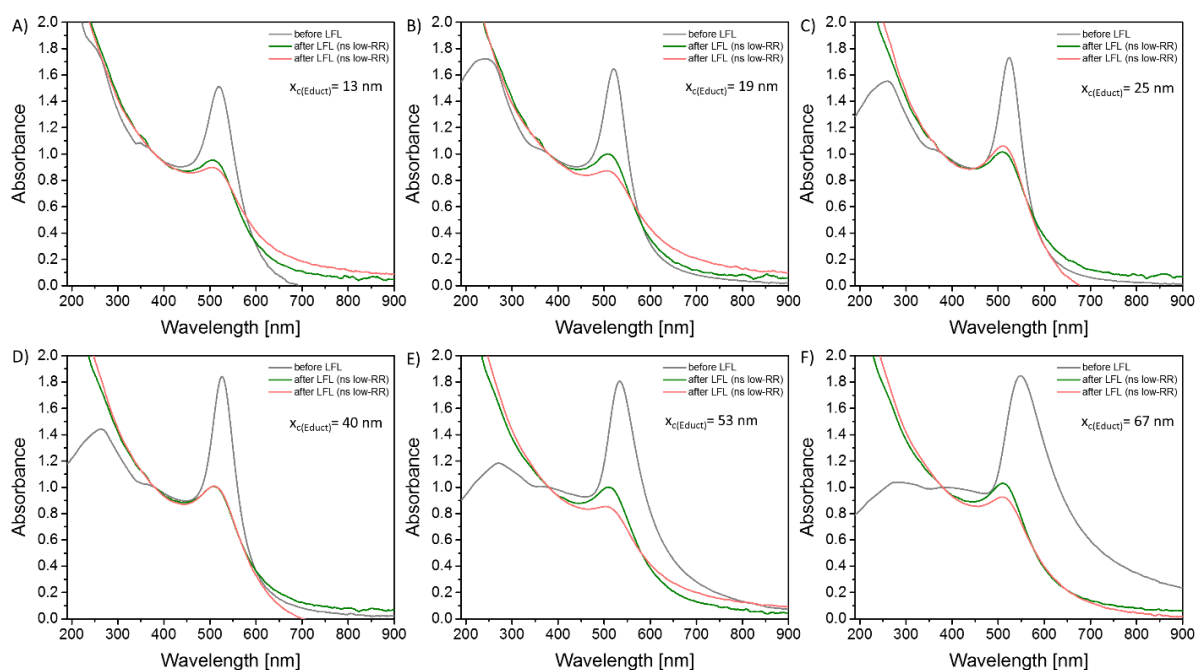


Figure S 6: Absorbance spectra (before and after LFL) in relation to the educt particle size:

A) 13 nm, B) 19 nm, C) 25 nm, D) 40 nm, E) 53 nm, and F) 67 nm.

Influence of the number of pulses per volume

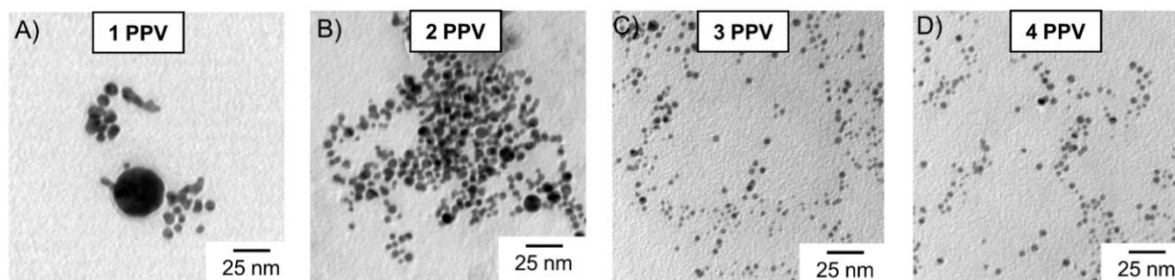


Figure S 7: TEM Images after LFL with 1–4 pulses per volume (PPV). A) 1 PPV, B) 2 PPVs, C) 3 PPVs, and D) 4 PPVs

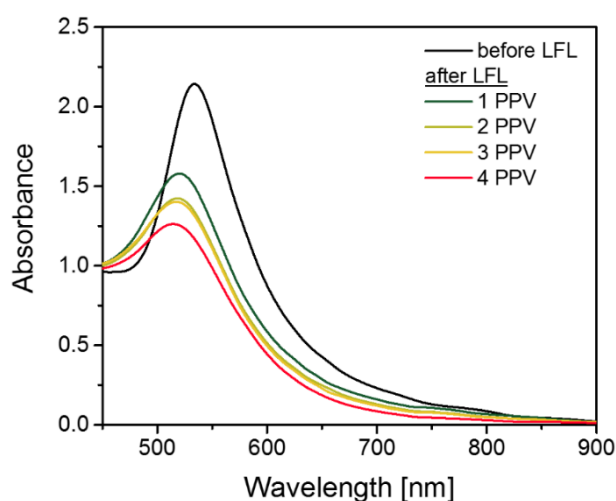


Figure S 8: Comparative UV/Vis-spectra of particles synthesized with a different number of PPVs. The spectra show a gradual blue shift and decrease in surface plasmon resonance (SPR) peak intensity, which confirms the smaller particle formation from 0 to 4 PPVs

Stability experiment against salt concentrations

We checked the colloidal stability of these nanoparticles synthesized via a different number of pulses per volume against the sodium chloride salt concentration. To do so, we titrated suspensions of AuNPs corresponding to different passages against increasing concentrations of

NaCl (0 – 1250 mM) and evaluated their aggregation states (measured after 24 h) by measuring their hydrodynamic diameters (d_h) after 24 h. We determined the hydrodynamic diameters of the nanoparticle suspensions (Figure S9) using the particle analyzer Malvern Zetasizer Nano ZS (Malvern Instruments Limited, UK) at 25 °C. We collected three sets of this data and used the average value in Figures 5 B and 5 D. To better evaluate the stability against salt, in these figures, we show only the changes in the hydrodynamic diameter after the addition of the corresponding aliquot of NaCl.

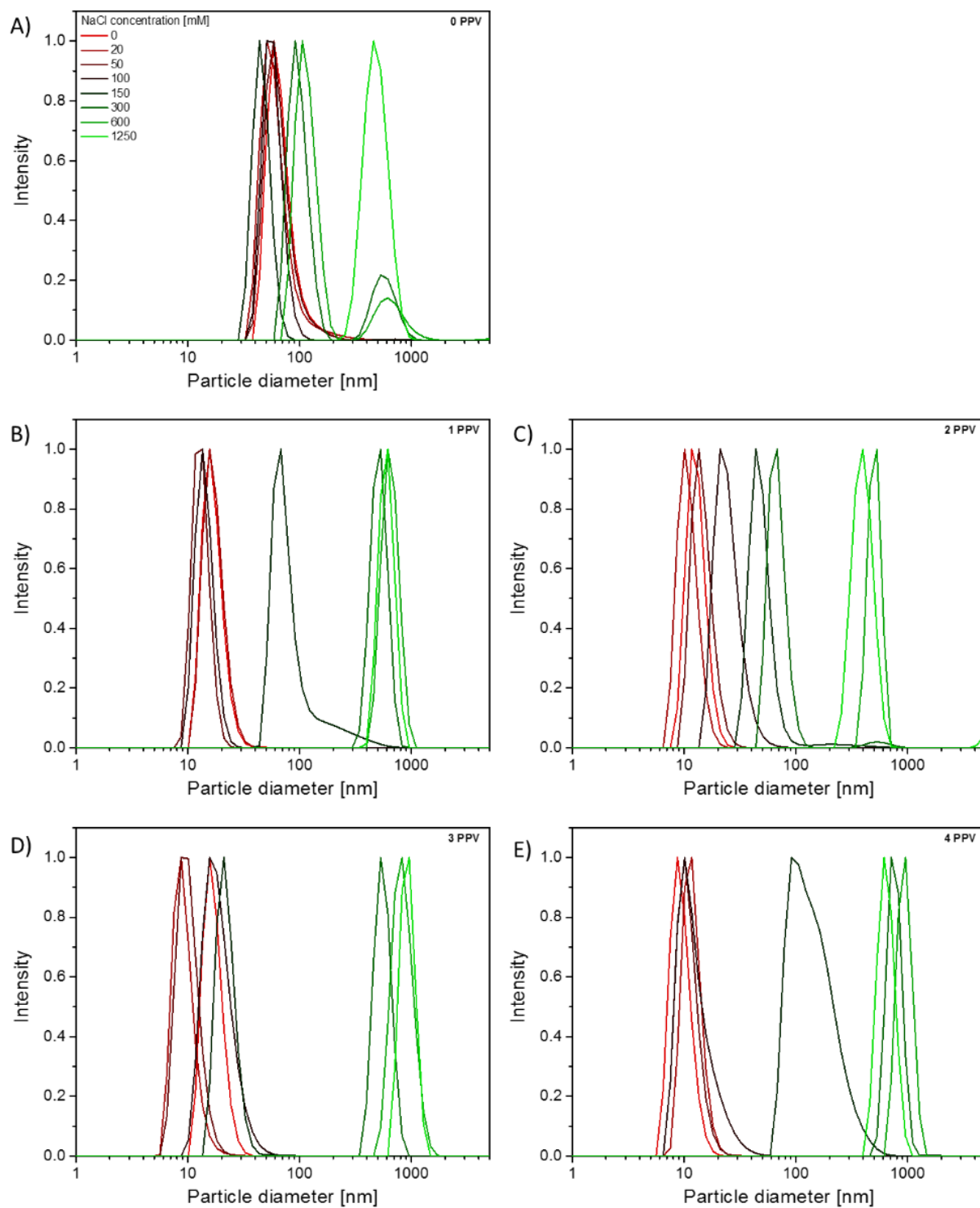


Figure S9: Size distribution plot of AuNPs after 1 to 4 PPVs in water in the presence of different concentrations of NaCl (recorded after 24 h). The NaCl concentrations ranged from 0.0 M to 1.25 M, as illustrated in the legend.

Experimental observation of the laser intensity

To determine the laser stability, we recorded the laser power over a period of one hour recording the power every 10 ms. Figure S 10 shows the experimentally determined stability of the laser 'ns-high-RR' (the data for laser 'ns-low-RR' are comparable, data not shown). If the laser stability is considered over 60 minutes (long-term) the deviation from the mean value is 1.64% for 'ns-high-RR' (1.40% for laser 'ns-low-RR'). If the laser intensity in the smallest possible measuring range is considered (10 ms), a deviation of 1.19% can be observed for 'ns-high-RR' (1.15% for 'ns-low-RR'). The short-term deviation of the laser intensity is plotted as error bars of the x-axis in Figure 6.

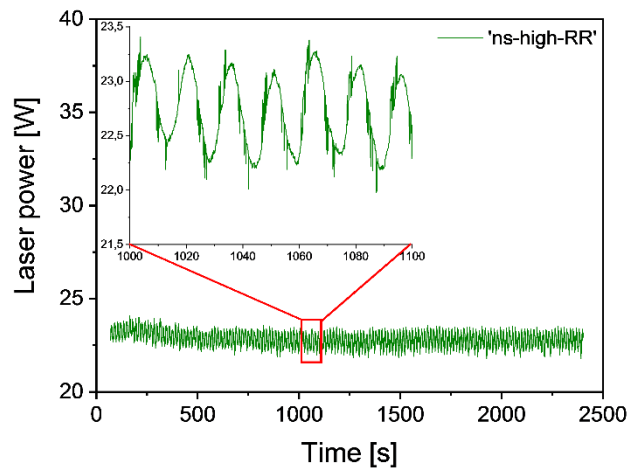


Figure S 10: Long-term stability of the laser power for 'ns-high-RR.'

Influence of pulse intensity on product particle size.

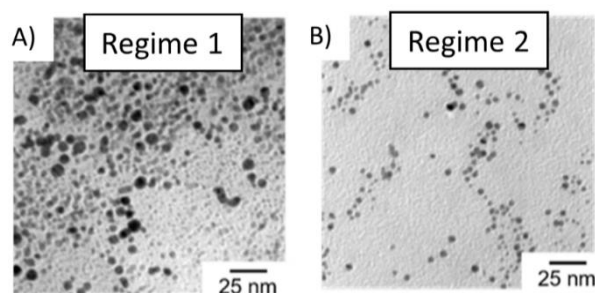


Figure S 11: TEM Images after fragmentation in regimes 1 (A) and 2 (B).

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